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Next item →

1. Which of the following is a necessary and sufficient condition for a function, f, to be continuous at the point $x = 4$? 1 / 1 point

- (●) $\lim_{x \to 4} f(x) = f(4)$.
- ( ) f is defined on an open interval that contains $x = 4$.
- ( ) $\lim_{x \to 4} f(x)$ exists.
- ( ) $\lim_{x \to 4} f(x)$ does not exist.

Nice work

This equation shows that all three conditions for continuity are satisfied.

2. Consider the function: 1 / 1 point

$$f(x) = \frac{x^2 - 1}{x - 1}$$

Which of the conditions of continuity are **not** met by $f(x)$ at $x = 1$?

- [x] $f(1)$ must be defined.

Nice work

Plugging $x = 1$ into the expression for $f(x)$ results dividing by 0, so $f(1)$ is not defined.

- [ ] $\lim_{x \to 1} f(x)$ must exist.
- [x] $\lim_{x \to 1} f(x) = f(1)$.

Nice work

Even though the limit exists, $f(1)$ is undefined and therefore the limit and function value are not equal.

3. Let $f(x) = \frac{x^2 - 9}{x - 3}$. 1 / 1 point

Is $f(x)$ continuous at $x = 3$?

- ( ) Yes, $f(x)$ is continuous at $x = 3$.
- (●) No, $f(x)$ is not continuous at $x = 3$.

Nice work

$\lim_{x \to 3} f(x)$ exists, and can be evaluated with factoring. However, $f(3)$ is undefined. Therefore, $f(x)$ has a removable discontinuity at $x = 3$.

4. Suppose $f(x) = \frac{x^2 - 4x + 3}{x^2 - 5x + 6}$. 1 / 1 point

Which statement describes the continuity of $f(x)$ at $x = 3$?

- ( ) f is continuous at $x = 3$.
- (●) f has a removable discontinuity at $x = 3$.
- ( ) f has a jump (nonremovable) discontinuity at $x = 3$.
- ( ) f has an infinite discontinuity at $x = 3$.

Nice work

Note that the function factors as $f(x) = \frac{x^2 - 4x + 3}{x^2 - 5x + 6} = \frac{(x-3)(x-1)}{(x-3)(x-2)}$.

Evaluating at $x = 3$ results in dividing by zero, so $f(3)$ is undefined.

However, the limit exists and can be found by canceling common factors:

$$\lim_{x \to 3} f(x) = \lim_{x \to 3} \frac{x - 1}{x - 2} = \frac{2}{1} = 2$$

Therefore, $f(x)$ has a removable discontinuity at $x = 3$.

5. Suppose $f(x) = \begin{cases} x^2 + 7, & x < 0 \\ x + 7, & x > 0 \end{cases}$ 1 / 1 point

Which statement describes the continuity of f at $x = 0$?

- ( ) f is continuous at $x = 0$.
- (●) f has a removable discontinuity at $x = 0$.
- ( ) f has a jump (nonremovable) discontinuity at $x = 0$.
- ( ) f has an infinite discontinuity at $x = 0$.

Nice work

First, note that 0 is not in the domain of f, so $f(0)$ is undefined.

We can find $\lim_{x \to 0} f(x)$ by finding the left and right limits with the piecewise definition.

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0} (x^2 + 7) = 7$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0} (x + 7) = 7$$

Therefore, $\lim_{x \to 0} f(x) = 7$.

Since the limit is defined while $f(0)$ is undefined, the function has a removable discontinuity at $x = 0$.

6. Let $f(x) = \frac{1}{x + 2}$ for $x \neq -2$ and $f(x) = 1$ for $x = -2$. 1 / 1 point

Explain why the function is discontinuous at -2.

- ( ) $f(-2)$ is not defined
- (●) $f(-2)$ is defined, but $\lim_{x \to -2} f(x)$ does not exist.
- ( ) $f(-2)$ and $\lim_{x \to -2} f(x)$ are both defined, but $\lim_{x \to -2} f(x) \neq f(-2)$.
- ( ) $\lim_{x \to -2^-} f(x)$ and $\lim_{x \to -2^+} f(x)$ are both defined, but $\lim_{x \to -2^-} f(x) \neq \lim_{x \to -2^+} f(x)$

Nice work

For this function, $f(-2)$ is defined and equals 1.

However, $\lim_{x \to -2} f(x)$ does not exist.

Note that both the lefthand and righthand limits at $x = -2$ are infinite, so neither is defined.

7. Let $f(x) = \sin(x)$ if $x < \pi/4$ and $f(x) = \cos(x)$ if $x \geq \pi/4$. 1 / 1 point

Explain why the function is continuous on $(-\infty, \infty)$.

- ( ) f is continuous on $(-\infty, \infty)$ because both $\sin(x)$ and $\cos(x)$ are continuous on $(-\infty, \infty)$, and the sum of continuous functions is continuous.
- ( ) f is continuous on $(-\infty, \infty)$ because both $\sin(x)$ and $\cos(x)$ are continuous on $(-\infty, \infty)$, and the composition of continuous functions is continuous.
- (●) f is continuous on $(-\infty, \infty)$ because both $\sin(x)$ and $\cos(x)$ are continuous on $(-\infty, \infty)$, and $\sin(x) = \cos(x)$ when $x = \pi/4$.
- ( ) f is continuous on $(-\infty, \infty)$ because both $\sin(x)$ and $\cos(x)$ are continuous on $(-\infty, \infty)$, and $\sin(x) \leq \cos(x)$ when $x \leq \pi/4$.

Nice work

Both $\sin(x)$ and $\cos(x)$ are continuous on $(-\infty, \infty)$.

Thus the only possible point of discontinuity is $\pi/4$ where f changes definition.

However, since $\sin(\pi/4) = \cos(\pi/4)$, the left- and right-hand limits agree with each other and with the function value, and thus f is continuous at $\pi/4$.

8. Let $f(x) = \frac{1}{1 + e^{1/x}}$. 1 / 1 point

Select all of the points below where $f(x)$ is discontinuous.

- [x] $x = 1$
- [ ] $x = -1$
- [x] $x = 0$

Nice work

At $x = 0$, the exponent $\frac{1}{x}$ is undefined. Therefore $f(0)$ is undefined, so $f(x)$ is not continuous at $x = 0$.

- [ ] $f(x)$ is continuous everywhere.

9. $f(x) = \begin{cases} 2^x, & \text{if } x \leq 1 \\ 3 - x, & \text{if } 1 < x \leq 4 \\ \sqrt{x}, & \text{if } x > 4 \end{cases}$ 0 / 1 point

Select all of the points below where $f(x)$ is discontinuous.

- [ ] $x = 4$
- [x] $x = 0$

This should not be selected

Each of the three functions that makes up the piecewise function is continuous on its domain. In particular, near $x = 0$, $f(x) = 2^x$, which is continuous everywhere.

Thus the possible points of discontinuity are those where the pieces meet: $x = 1$ and $x = 4$. Since the function is defined at both of these points, we must check that it is equal to the limit.

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} 2^x = 2$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1} (3 - x) = 2$$

Therefore $\lim_{x \to 1} f(x) = 2 = f(1)$, so f is continuous at $x = 1$.

$$\lim_{x \to 4^-} f(x) = \lim_{x \to 4} (3 - x) = -1$$

$$\lim_{x \to 4^+} f(x) = \lim_{x \to 4} \sqrt{x = 2}$$

Therefore, $\lim_{x \to 4} f(x)$ does not exist, so f is discontinuous at $x = 4$.

- [ ] $f(x)$ is continuous everywhere
- [x] $x = 1$

This should not be selected

Each of the three functions that makes up the piecewise function is continuous on its domain. Thus the possible points of discontinuity are those where the pieces meet: $x = 1$ and $x = 4$. Since the function is defined at both of these points, we must check that it is equal to the limit.

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} 2^x = 2$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1} (3 - x) = 2$$

Therefore $\lim_{x \to 1} f(x) = 2 = f(1)$, so f is continuous at $x = 1$.

$$\lim_{x \to 4^-} f(x) = \lim_{x \to 4} (3 - x) = -1$$

$$\lim_{x \to 4^+} f(x) = \lim_{x \to 4} \sqrt{x = 2}$$

Therefore, $\lim_{x \to 4} f(x)$ does not exist, so f is discontinuous at $x = 4$.

10. Find the value of the constant c that makes the function f continuous everywhere. 1 / 1 point